

codebase-memory-mcp: Map a Whole Codebase So AI Agents Use 99% Fewer Tokens

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DeusData released [codebase-memory-mcp](#), an open-source tool that indexes a whole repository into a knowledge graph: a map of how functions, files, and packages connect, so the agent queries that map instead of reading files one by one. It plugs into coding assistants through

[MCP](#), the open standard that lets a model call outside tools, and ships as a single binary with no Docker, no API key, and no setup. It reads 158 languages, indexed the 28-million-line Linux kernel in 3 minutes, and answers most lookups in under a millisecond. In a [paper](#) testing 31 real repositories, it reached 83% answer quality against 92% for an agent reading files, at 10 times fewer tokens and 2.1 times fewer tool calls.

Tokens are the hidden bill on every agent task, and crawling a repo is where most get spent. One graph query replaces dozens of search-and-read cycles: on a five-question test, that cut usage from 412,000 tokens to 3,400, a 99% drop. The tool runs fully on your machine, stores nothing, and installs into 11 agents like Claude Code and Codex with one command. The trade is honest, a little less accuracy for a fraction of the cost.

Agents are shifting from reading code like prose to querying it like a database. We previously covered [CodeGraph](#), which trimmed tokens 47% the same way; this pushes the idea into one small, language-agnostic binary that any tool can share.

Sources:

- [DeusData/codebase-memory-mcp on GitHub](#)
- [Codebase-Memory: Tree-Sitter-Based Knowledge Graphs for LLM Code Exploration via MCP \(arXiv\)](#)
- [Paper HTML version](#)
- [Model Context Protocol](#)

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